

Atomic Multiplier Charts and BCP for Fell Algebras with Separable Irreps

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Definition 1. *A C^* -algebra A has the ball-covering property, or BCP, if the unit sphere of A is covered by countably many open balls none of which contains the origin. It has the uniform ball-covering property, or UBCP, if the balls can be chosen with a common positive gap from the origin.*

For an open set $O \subset \text{Prim}(A)$, let A_O denote the corresponding closed two-sided ideal of A .

Statement

Theorem 1 (atomic multiplier chart criterion). *Let A be a C^* -algebra. Assume that $\text{Prim}(A)$ has a countable π -basis (O_n) such that every patch is trivial continuous trace, that is,*

$$A_{O_n} \cong C_0(O_n, \mathcal{K}(H_n))$$

for a separable nonzero Hilbert space H_n . Then A has UBCP.

Corollary 1 (Fell algebras with separable irreducible representations). *Let A be a Fell C^* -algebra all of whose irreducible representations act on separable Hilbert spaces. Then*

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(A) \text{ has a countable } \pi\text{-basis.}$$

Idea of the Proof

The earlier compact-chart packet required every ambient element of A to be represented on a chart by a norm-continuous compact-operator field. That is stronger than necessary. On a continuous-trace patch

$$A_O \cong C_0(O, \mathcal{K}(H)),$$

an ambient element $a \in A$ acts as a multiplier, hence as a bounded strict-continuous $B(H)$ -field $P \mapsto a(P)$. The whole field need not be norm-continuous, but every fixed matrix coefficient

$$P \mapsto \langle a(P)\xi, \eta \rangle$$

is continuous.

Given $a \in S_A$, choose a primitive point where $\|a(P)\| > 3/4$. On a continuous-trace patch around that point, choose unit vectors ξ, η from a countable dense set so that

$$\text{Re}\langle a(P)\xi, \eta \rangle > \theta$$

for a fixed $\theta > 0$. By coefficient continuity this remains true on a smaller open set. A scalar bump h supported there gives a centre

$$c = Mh(\eta \otimes \xi^*).$$

The operator estimate below says that a stays within a radius $\rho < M$ of this centre at every primitive quotient. Since $\|c\| = M$, the balls are uniformly off the origin.

Proof

Lemma 1 (rank-one coefficient estimate). *Fix $0 < \theta < 1$. There are constants $M > 1$ and $\rho < M$ such that the following holds for every Hilbert space H . If $T \in B(H)$ satisfies $\|T\| \leq 1$, if $\xi, \eta \in H$ are unit vectors, and if*

$$\text{Re}\langle T\xi, \eta \rangle \geq \theta,$$

then, for the rank-one operator $Ru = \langle u, \xi \rangle \eta$,

$$\sup_{0 \leq t \leq M} \|T - tR\| \leq \rho.$$

Proof. Choose $0 < \varepsilon < 1$ so small that

$$2\theta(1 - \varepsilon) - 2\sqrt{\varepsilon} \geq \theta.$$

Put $\delta = \theta/8$. We first show that for all sufficiently large t ,

$$\|T - tR\| \leq t - 2\delta.$$

Let $x \in H$ be a unit vector, and write

$$x = \lambda \xi + v, \quad v \perp \xi, \quad s = |\lambda|.$$

Then $\|v\| = (1 - s^2)^{1/2}$. Since $\|T\| \leq 1$,

$$\begin{aligned} \|(T - tR)x\|^2 &= \|Tx\|^2 + t^2 s^2 - 2t \operatorname{Re}(\bar{\lambda} \langle Tx, \eta \rangle) \\ &\leq 1 + t^2 s^2 - 2t\theta s^2 + 2ts(1 - s^2)^{1/2}. \end{aligned}$$

If $s^2 \leq 1 - \varepsilon$, then

$$\|(T - tR)x\|^2 \leq t^2(1 - \varepsilon) + 2t + 1,$$

which is at most $(t - 2\delta)^2$ for all sufficiently large t . If $s^2 > 1 - \varepsilon$, then the choice of ε gives

$$-2\theta s^2 + 2s(1 - s^2)^{1/2} \leq -\theta,$$

and therefore

$$\|(T - tR)x\|^2 \leq t^2 - \theta t + 1.$$

This is also at most $(t - 2\delta)^2$ for all sufficiently large t . Thus there is t_0 such that $\|T - tR\| \leq t - 2\delta$ whenever $t \geq t_0$.

Choose $M > t_0 + 1 + 2\delta$, and set $\rho = M - \delta$. If $0 \leq t < t_0$, then

$$\|T - tR\| \leq 1 + t < t_0 + 1 < M - 2\delta < \rho.$$

If $t_0 \leq t \leq M$, then

$$\|T - tR\| \leq t - 2\delta \leq M - 2\delta < \rho.$$

Hence $\sup_{0 \leq t \leq M} \|T - tR\| \leq \rho < M$. □

Lemma 2 (countable atomic bump data). *Under the hypotheses of Theorem 1, one can choose, for each chart O_n :*

1. *an isomorphism*

$$\Phi_n : A_{O_n} \longrightarrow C_0(O_n, \mathcal{K}(H_n));$$

2. *a countable dense set D_n in the unit sphere of H_n ;*

3. *a countable family $\mathcal{B}_n \subset C_c(O_n)$, with $0 \leq h \leq 1$ and $\|h\| = 1$, such that every nonempty open subset of O_n contains the support of some $h \in \mathcal{B}_n$.*

Proof. The Hilbert spaces H_n are separable, so the sets D_n exist. The continuous-trace trivializations are part of the hypothesis, after refining the countable π -basis if necessary. Since O_n is locally compact Hausdorff and the O_m 's form a π -basis, local regularity gives countably many compactly supported scalar bumps with cofinal supports. □

Proof of Theorem 1. Fix $\theta = 1/2$, and let M and $\rho < M$ be the constants supplied by Lemma 1. Put

$$r = \frac{M + \rho}{2},$$

so that $\rho < r < M$. For every chart O_n , every $h \in \mathcal{B}_n$, and every $\xi, \eta \in D_n$, define

$$c(n, h, \xi, \eta) = \Phi_n^{-1}(Mh(\eta \otimes \xi^*)) \in A_{O_n} \subset A.$$

These centres are countable and have norm M . We claim that the balls

$$B(c(n, h, \xi, \eta), r)$$

cover S_A . Since $r < M$, this proves UBCP.

Let $a \in S_A$. The set

$$W = \{P \in \text{Prim}(A) : \|a + P\| > 3/4\}$$

is nonempty and open. Choose a chart $O_n \subset W$, and then choose $P_0 \in O_n$. Let $T_{P_0} \in B(H_n)$ be the operator representing the multiplier action of a on the fibre over P_0 . Then

$$\|T_{P_0}\| = \|a + P_0\| > 3/4.$$

Choose unit vectors $\xi_0, \eta_0 \in H_n$ such that

$$\text{Re}\langle T_{P_0}\xi_0, \eta_0 \rangle > 2/3.$$

After approximating ξ_0, η_0 by vectors in the dense set D_n , we may choose $\xi, \eta \in D_n$ with

$$\text{Re}\langle T_{P_0}\xi, \eta \rangle > 1/2.$$

The canonical action of A on $A_{O_n} \cong C_0(O_n, \mathcal{K}(H_n))$ is a multiplier field. In this trivialization, multiplier fields are bounded strict-continuous $B(H_n)$ -valued fields; hence the scalar coefficient

$$P \mapsto \text{Re}\langle T_P\xi, \eta \rangle$$

is continuous. Therefore there is a nonempty open neighbourhood

$$N \subset O_n$$

of P_0 such that

$$\text{Re}\langle T_P\xi, \eta \rangle \geq 1/2 \quad (P \in N).$$

Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$, and set

$$c = c(n, h, \xi, \eta).$$

If $P \in \text{supp } h$, then in the fibre over P ,

$$c(P) = Mh(P)(\eta \otimes \xi^*),$$

and Lemma 1, applied with $t = Mh(P)$, gives

$$\|a + P - (c + P)\| \leq \rho < r.$$

If $P \in O_n \setminus \text{supp } h$, then $c(P) = 0$, so

$$\|a + P - (c + P)\| = \|a + P\| \leq 1 < r,$$

after increasing M , if necessary, so that $\rho > 1$. If $P \notin O_n$, then $c \in A_{O_n} \subset P$, and again

$$\|a + P - (c + P)\| = \|a + P\| \leq 1 < r.$$

Taking the supremum over primitive quotients gives $\|a - c\| < r$. Thus the displayed countable family of balls covers S_A , and A has UBCP. $\square$

Proof of Corollary 1. If A has BCP, the primitive π -base obstruction from the earlier lane-19 packet gives a countable π -basis for $\text{Prim}(A)$. UBCP always implies BCP.

It remains to prove that a Fell algebra with separable irreducible representations and countable primitive π -basis satisfies the hypothesis of Theorem 1. Fell spectra are locally Hausdorff. Given a nonempty open set $U \subset \text{Prim}(A)$, choose a nonempty open Hausdorff subset $V \subset U$. The ideal A_V is Fell with Hausdorff spectrum, hence has continuous trace. Local triviality gives a nonempty open

$$O \subset V$$

such that

$$A_O \cong C_0(O, \mathcal{K}(H))$$

for the Hilbert space of the irreducible representations over that patch. By assumption H is separable. Now start from any countable π -basis $\mathcal{P}$ of $\text{Prim}(A)$, and retain those members of $\mathcal{P}$ which are contained in one of the continuous-trace trivialization patches just constructed. The preceding paragraph shows that this retained subfamily is still a countable π -basis, and each retained member inherits the corresponding separable continuous-trace trivialization. Thus the hypotheses of Theorem 1 hold. Therefore A has UBCP. $\square$

Scope and Limitations

This packet settles the positive Fell case under separability of irreducible representations, but it does not claim a full arbitrary C^* -classification. The remaining type-I question is whether countable primitive π -base forces enough continuous-trace open patches outside the Fell class, or whether there are non-Fell CCR/type-I algebras with countable primitive π -base and separable irreducible representations that still fail BCP.

The separability assumption on irreducible representations is also real for this proof: it supplies the countable rank-one directions. Earlier lane-19 packets show that nonseparable compact-operator fibres can fail BCP even with one-point primitive spectrum.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, Fell algebras, separable irreducible representations, local continuous-trace patches, atomic multiplier charts, BCP, UBCP, and primitive π -bases. Web searches on June 17, 2026 for Fell algebras, locally Hausdorff spectrum, continuous trace patches, and arXiv:1207.2540 located the Clark–an Huef–Raeburn Fell reference [2], but no BCP theorem of the above form. The result is therefore recorded as an agent-produced partial theorem, subject to human literature review.

Human Review Focus

The core of the proof is Lemma 1. Reviewers should also check:

1. strict multiplier continuity implies continuity of fixed scalar matrix coefficients on a continuous-trace trivialization;
2. the countable bump/rank-one parameter selection;
3. the Fell-to-continuous-trace patch reduction in the corollary.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Lisa Orloff Clark, Astrid an Huef, and Iain Raeburn, *The equivalence relations of local homeomorphisms and Fell algebras*, arXiv:1207.2540, 2012.
- [3] J. Dixmier, *C*-Algebras*, North-Holland, 1977.
- [4] I. Raeburn and D. P. Williams, *Morita Equivalence and Continuous-Trace C*-Algebras*, American Mathematical Society, 1998.
- [5] Run `fa_banach_001`, lane 19, *A Primitive-Space π -Base Obstruction for C*-Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
- [6] Run `fa_banach_001`, lane 19, *Nonseparable Compact Operators Fail BCP*, solution packet `2311.14257_nonseparable_compact_operators_fail_bcp`, 2026.